



YOUR KINDLE NOTES FOR:

Carrying the Fire: 50th Anniversary Edition

by Michael Collins

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144 Highlights | 2 Notes

Highlight (Yellow) | Location 36

It is easy to so identify with the story that you become an astronaut yourself.

Highlight (Yellow) | Location 38

On the porch of his home at Worcester, Massachusetts, in 1929, I listened to Goddard talk about his liquid rockets and his dreams of spatial exploration. He felt quite certain, he told me, that he would be able to build a rocket capable of carrying scientific instruments to altitudes approaching one hundred miles; and that from these altitudes, above atmospheric interference, valuable observations could be made. He thought it would even be possible to build a multistage rocket that could reach the moon; but “it might cost a million dollars,” he added, with an air of dismissing the idea from consideration. At that time, none of Goddard’s rockets had reached an altitude of one mile, to say nothing of one hundred. His estimated budget for a year’s experimental work was \$25,000, including laboratory, launching tower, equipment, salaries, and transportation. Now I was reading about an accomplished voyage to the moon, and the United States was spending millions of dollars every day on the development of rockets.

Highlight (Yellow) | Location 46

Carrying the Fire roots into Goddard’s pioneering work, and branches in modern technology’s accomplishments. You must have witnessed an Apollo ascent to know the title’s aptness. When a countdown at Cape Canaveral touches zero, you think the giant rocket will be consumed by bursting clouds of flame and thunder. Where I stood with astronauts three miles away from Apollo 11’s pad, my chest was beaten and the ground shook as though bombs were falling nearby. Then a flame arose, left the ground behind—higher—faster—a meteor streaking through the sky. It seemed impossible for life to exist while carrying that ball of fire. Yet it did exist, as I saw later on a television screen that portrayed weightless men, countless instruments, and planet earth through a window.

Highlight (Yellow) | Location 55

How isolated, how lonely those two space supermen appeared! But they had each other for companionship; and through television, they were held in the thoughts of viewing millions of men and women. To be really isolated, to fully experience loneliness, you must be alone. From Armstrong’s and Aldrin’s spectacular movements, my mind shifted to Collins’s lunar orbiting. Relatively inactive and unwatched, he had time for contemplation,

Highlight (Yellow) | Location 59

Here was human awareness floating through universal reaches, attached to our earth by such tenuous bonds as radio waves and star sights. A minor functional error would leave it floating forever in the space from which, ancestrally, it came. Only once before had I felt such extension as when I thought of Astronaut Collins. That was over the Atlantic Ocean on my nonstop flight with the Spirit of St. Louis.

Highlight (Yellow) | Location 62

I had been without sleep for more than two days and two nights, and my awareness seemed to be abandoning my body to expand on stellar scales. There were moments when I seemed so disconnected from the world, my plane, my mind and heartbeat that they were completely unessential to my new existence. Experiences of that flight combined with those of ensuing life have caused me to value all human accomplishments by their effect on the intangible quality we name "awareness."

Highlight (Yellow) | Location 67

After evolutionary epochs, life in the human species became aware of its awareness. Then in a comparatively short period of time, using such devices as lenses and spacecraft for assistance, man has extended his awareness outwardly through the universe and inwardly through the atom.

Highlight (Yellow) | Location 70

We find the speed of light and the vastness of space to be incompatible with biological time.

Highlight (Yellow) | Location 75

We see life compressed on a shrunken orb at the same time we feel awareness expanded to fantastically great dimensions. Can we harmonize the two? What does the future hold for man?

Highlight (Yellow) | Location 87

Is it remotely possible that we are approaching a stage in evolution when we can discover how to separate ourselves entirely from earthly life, to abandon our physical frameworks in order to extend both inwardly and outwardly through limitless dimensions of awareness? In future universal explorations, may we have no need for vehicles or matter?

Highlight (Yellow) | Location 115

Mars takes a gigantic loop around the sun, as do we. Our elliptical orbits are such that we come as close as 35 million miles to each other and as far as 220 million. A rocket leaving Earth for Mars has a choice of many possible paths, but the most economical in terms of fuel is called a Hohmann transfer, and takes between six and nine months one way. Our two orbits are such, however, that a crew arriving at Mars from a Hohmann would have to wait a year or more for a favorable return alignment. Therefore, a round trip would take more than two years.

Highlight (Yellow) | Location 122

As for the environment, probably the worst problem is radiation, both solar and cosmic—high-speed particles from the far corners of the universe. As survivors of Hiroshima can attest, radiation can cause terrible damage to the human body. Even shielded from it, the crew would have to adjust to weightlessness if their craft was not producing its own gravity, probably a lot less than one G if it is rotating. Our experience with the space station shows that cardiovascular problems can ensue, and that reduced G can cause increased pressure on the eyes, from within, that can damage vision.

Highlight (Yellow) | Location 127

Once on Mars, with about a third of one Earth G, the crew should be quite comfortable (perhaps even exuberant), but must still be provided with sufficient oxygen and protected from radiation. The atmosphere is too thin to be of much help, and is quite toxic as well. All in all, Apollo was child's play compared to a Mars landing.

Highlight (Yellow) | Location 139

The moon's jagged contours certainly stick in my memory but just barely, compared with the vision that I summon over and over again of the itsy-bitsy sphere just outside my window, motionless, cradled in black velvet. That sight—the Earth, tiny, shiny, blue of sky and water, white of clouds, with only a brown trace of land—haunts me. I can blot it out with my thumb, but back it comes, 200,000 miles away, quiet and peaceful-looking even though I know that it is neither. Surprise! If allowed only one word to describe Earth, I would say fragile. Give me a second choice and I'll pick inhabited. Now, that is a reach, because that is what I could not see. But it doesn't matter, because I did feel something—a presence—like tiny creatures crawling ant-like over Earth's surface.

Highlight (Yellow) | Location 153

it took the biologist James Lovelock to choose the name Gaia as the sum total of all earthly life-forms. I guess he meant all the way, from E. coli and elephants to

Highlight (Yellow) | Location 155

We lunar crews looked back at 3 billion earthlings. Almost fifty years later, the number is about 8 billion, ever increasing, with predictions of 10 billion or more by mid-century. This growth alarms some people, but not too many, I think;

Highlight (Yellow) | Location 157

I tried to condense the situation in my concluding remarks in the Preface to this book's 2009 edition: "We need a new economic paradigm to produce prosperity without growth." I believe this even more firmly today. I believe Gaia is saying "Ouch!" The increased burden on our resources is being absorbed, but at a price: more CO₂ in the atmosphere, warmer temperatures, rising seas. The oceans are getting a double whammy, becoming warmer and

more acidic. CO2 may be good for our forests, but my favorite waters, surrounding the Florida Keys, are experiencing widespread coral disease, whitening and dying due to these factors. Australia's Great Barrier Reef is similarly afflicted.

Highlight (Yellow) | Location 198

In the sciences, thermodynamics was my favorite, if only dimly grasped. Entropy and enthalpy seemed pretty cool. In the humanities, it was Chinese history, the wild, diverse, colorful, exotic dynasties somehow winning out over Washington and Jefferson, whose lives seemed more subdued and conformist. At West Point we had an exhortation: "When the going gets tough, the tough get going." When my going toughened, I joined the Literary Guild and purchased their monthly book, usually a novel. My roommates, sweating through differential equations, thought I was crazy, but it was just my extremely mild form of rebellion.

Highlight (Yellow) | Location 216

The Massachusetts Institute of Technology is my favorite of the technical institutions, probably because I used to go there in the 1960s to learn the inner mysteries of the Apollo guidance system, which they designed. Now they invite this old codger back to lecture, which is ridiculous, sort of like asking a shower singer to serenade the divas at the Met.

Highlight (Yellow) | Location 220

Focus on STEM—science, technology, engineering, and math. Thus they will become educated, they are led to believe. I say no. STEM may be a beginning, but it's far from a complete education. Perhaps that is because I have worked with too many incoherent engineers. I want to change STEM to STEEM, inserting English. Others, including Buzz Aldrin, have pushed for a place for the arts—i.e., STEAM—but I say no to that also. I am a watercolor painter, but that is a frill. If I botch a painting, it's a private failing. But someone who mangles the English language in public, that's altogether different. More common is the engineer who has an important point to make, verbally or in a memo, but cannot really get it across, as it is immersed in a cloud of jargon.

Highlight (Yellow) | Location 226

I try to use poetry as an antidote. Don't concentrate so much on the rules, but see how others have used our beautiful language.

Highlight (Yellow) | Location 227

usually end up reciting, in a loud voice and by memory, a long passage from John Milton's Paradise Lost.

Highlight (Yellow) | Location 230

My twenty-plus years in retirement have been fun and fulfilling in many ways. Watching my family grow, all doing well, has been paramount, but beyond that come fishing, reading, chasing the stock market, painting, and exercise, exercise, exercise. I'm not wealthy, but I have plenty to enjoy all these things in comfortable, or maybe even luxurious, surroundings ... Lucky, lucky, lucky. Michael Collins August 2018

Highlight (Yellow) | Location 241

it was what we had signed on to do. It was not, in the words describing the Congressional Medal of Honor, “above and beyond the call of duty.” It was not heroism.

Highlight (Yellow) | Location 243

Then there are the celebrities. What a senseless, empty concept for someone to be, as my friend the great historian Daniel Boorstin put it, “known for his well-knownness.” How many live-ins, how many trips to rehab, maybe—wow—you could even get arrested! All this can catapult an attractive youngster to the front ranks of the media,

Highlight (Yellow) | Location 250

I used to imagine a gigantic grandfather clock in the sky, its pendulum stuck over to the left in the “too young” position—you couldn’t drink, drive, make money, or do anything really interesting. Then one night you went to bed and—voilà!—the next morning the pendulum had swung over to the right, to the “too old” side: over the hill, don’t try running, getting bald. Was there nothing in the middle? For me, the middle was the 1960s: a great family, test pilot, astronaut, what more could I want? Just lucky, the right place at the right time.

Highlight (Yellow) | Location 254

In the early days of NASA, Houston was a great place to work. There were thirty of us in the astronaut office, and we were given interesting responsibilities and a lot of latitude in fulfilling them. We were generalists in that we were expected to learn the workings of the entire array of machines and the plans for using them to get to the moon.

Highlight (Yellow) | Location 261

Today, my life in retirement is a simple but entertaining one. I split my time between Florida and Boston, doing some watercolor painting, and fishing a lot in both places. I have a tiny paddleboat from which I gather snook in the south and striped bass in the north, getting good exercise along the way. Every once in a while I look up at the moon, but not too often: been there, done that, as I hope I have explained properly in this book. But from my boat I keep wishing I could see Mars.

Highlight (Yellow) | Location 270

I wrote Mission to Mars. In it, I pondered what there is in us—genes, character, culture, spirit, ethos, whatever—that makes us look up into the night sky and become curious and restless. To go, to see, to touch, to smell, to learn, to understand: I think we are wanderers, and this migratory drive, this extraterrestrial imperative, will surely lead us to our neighbor Mars.

Highlight (Yellow) | Location 276

after giving Mars a great deal of affectionate thought, I realized T. S. Eliot understood it better than I: We shall not cease from exploration And the end of all our exploring Will be to arrive where we started And know the place for the first time.

Highlight (Yellow) | Location 280

When I looked back at Earth from the moon, if I could use only one word to describe the tiny thing, it would have been fragile.

Highlight (Yellow) | Location 283

I don't think this growth is healthy or sustainable, but our economic models are predicated on growth; they require it. Grow or die or maybe both.

Highlight (Yellow) | Location 286

We need a new economic paradigm to produce prosperity without growth.

Highlight (Yellow) | Location 291

after Apollo 11, people still don't have the vaguest idea of what it was like "up there," or what pre- and post-flight activities were necessary and how they affected the lives of those involved. I wrote this book to do that,

Highlight (Yellow) | Location 295

finally orbiting the moon while Armstrong and Aldrin walked on it. Although undeniably autobiographical, I do not see it as a me-me-me kind of thing, but rather an insider's factual and simple explanation of how the machines operated, who operated them, and what it was like living in an artificial, high-pressure environment. Being an astronaut was as exciting a job as anyone could ever have,

Highlight (Yellow) | Location 300

In a way I regret not having written the book earlier, because some of the story is not as fresh in my mind as I would like. On the other hand, waiting does have its advantages, because in the intervening years I have learned more about what fascinates the non-astronaut, and I have spent more time on these points and less on some of the arcane aspects that appeal mightily to pilots but not at all to lawyers or housewives. Hopefully, I have also gained some perspective in regard to how flying in space has changed my life. But above all, I am glad that I wrote it myself. No matter how good the ghost, I am convinced that a book loses realism when an interpreter stands between the storyteller and his audience.

Highlight (Yellow) | Location 311

Apollo was the god who carried the fiery sun across the sky in a chariot,

Highlight (Yellow) | Location 312

how would you carry fire? Carefully, that's how, with lots of planning and at considerable risk.

Highlight (Yellow) | Location 315

the trip deserves the telling.

Highlight (Yellow) | Location 324

learning to ride a flying machine; if you are looking for perfect safety you will do well to sit on a fence and watch the birds, but if you really wish to learn you must mount a machine and become acquainted with its tricks by actual trial. —Wilbur Wright, 1901

Highlight (Yellow) | Location 333

For Edwards was all these things and I had been accepted as a member of Class 60-C at the USAF Experimental Flight Test Pilot School, along with thirteen other exalted ones, mostly Americans (one Italian, one Dane, one Japanese), mostly hyperthyroid, superachieving first sons of superachievers. To this day, I am impressed by this group; I love them, they leer at me from my study wall. One of them has walked on the moon, two have circled it; two—two of the best—are dead.

Highlight (Yellow) | Location 376

acquainted with such stomach-ache producers as under-ripe mangoes and overripe coconuts.

Highlight (Yellow) | Location 378

In Puerto Rico, I also took my first airplane ride, in a small twin-engine amphibian, the Grumman Widgeon.

Highlight (Yellow) | Location 382

he relished telling how he had, in 1911 in the Philippines, taken his first airplane ride, in a Wright machine, sitting on the wing next to Frank Lahm, who was the second military pilot to be taught by the Wrights.

Highlight (Yellow) | Location 397

the Army as a more sensible career choice. Against this was the wonder of what the next fifty years might bring. It had been less than fifty years since the Wrights first flew, and already we were into the jet age.

Highlight (Yellow) | Location 411

the sole channel into the two Fighter Wings in Korea, which, with their North American F-86 Sabrejets, were battling the MIGs so successfully. At Nellis we really learned to fly—a concentrated, aggressive course designed

to weed out anyone who might be a marginal performer in Korea. It was a brutal process as well. In the eleven weeks I was there, twenty-two people were killed. In retrospect it seems preposterous to endure such casualty rates without help from the enemy, but at the time the risk appeared perfectly acceptable.

Highlight (Yellow) | Location 418

At night we roared into Las Vegas, driving our cars in as close a formation as we flew our Sabrejets, terrorizing the natives,

Highlight (Yellow) | Location 420

It was a hectic time, and I'm surprised to have survived. I have never felt quite so threatened since.

Highlight (Yellow) and Note | Location 427

The trip had been fascinating (I have never seen anything from the air more beautiful than the clear azure blue of the Greenland glacier's fissured rim) and France was a new flying experience.

Only time i've been in the cockpit of a 747 in flight...

Highlight (Yellow) | Location 450

The school was dismal. The classroom work was stultifying; flying time was scarce and in obsolete equipment; classes started at six in the morning—what more can I say? In an effort to condense the misery somewhat, I started doubling up on my class load and finished the nine-month course in six months. Grand.

Highlight (Yellow) | Location 478

Although I had some nice recommendations, there was nothing in my background to make me believe I would be accepted. There must have been thousands like me. Therefore, when the fateful letter finally arrived, assigning me to "Class 60-C, USAF Experimental Flight Test Pilot School, commencing 29 August 60, course duration 32 weeks," I could not have been more pleased had I been offered a flight to the moon. Our crew of three (Pat and year-old Kate) moved the two hundred miles from Nellis to Edwards.

Highlight (Yellow) | Location 491

Tom Stafford, the chief of the Performance Section, was determined to get us off on the right foot. Doing the work of three himself, old Mumbles soon had us immersed in the intricacies of measuring the various performance parameters of new aircraft. The general pattern was: first, by classroom lectures, to learn the theoretical aspects of a particular type of test, then to make one or more flights trying out the new technique, and finally to analyze the reams of test data acquired during the flights. This last part, the reduction of thousands of bits of information ("data points") into an intelligible report, complete with charts and graphs, was a tiresome, time-consuming process.

Highlight (Yellow) | Location 509

We also had some memorable parties, usually on Friday night after a typical tedious week of academics, quizzes, a little flying, and a lot of data reduction. Then we would gather and drink and ignore our wives and discuss our future (after graduation) and complain about the few good jobs available and sometimes even sing, ditties whose words don't seem worth repeating but which generally described the tribulations of poor pilots in a non-comprehending world.

Highlight (Yellow) | Location 512

We also began to develop considerable confidence in our own abilities, and grudgingly had to admit that there was some merit to the test pilot school's approach, that the long hours were paying off. We learned to do things with airplanes which normally are not considered necessary or even possible.

Highlight (Yellow) | Location 521

the Air Force personnel system is not noted for its sensitivity or perspicacity.

Highlight (Yellow) | Location 523

the only available slot in Fighter Ops went, by what alchemy I will never know, to Mike Collins.

Highlight (Yellow) | Location 527

The most colorful racing pilot from the Golden Age of Aviation between the world wars, Roscoe had had us sitting goggle-eyed as he matter-of-factly described that wild world of aviation which we all knew was gone forever. Roscoe had flown with a waxed mustache and a pet lion named Gilmore; we flew with a rule book, a slide rule, and a computer. To get to the moon would require a very big computer, a whole basement full of computers, in fact.

Highlight (Yellow) | Location 531

in the spring of 1961 at Edwards, I was somewhere between Gilmore and moon, in the realm of Starfighters and Super Sabres and Delta Darts, sophisticated machines which bridged the gap between the colorful past I knew I had missed and the complex future I did not know was coming.

Highlight (Yellow) | Location 557

In fact, to me the amazing, the incredible, the fantastic (pardon the slip into astronaut parlance) fact is that so few crew mistakes and mishaps have occurred during Gemini and Apollo flights. The opportunities for error are almost limitless, and only superlative design permits virtually error-free operation of a machine as complex as an Apollo command module. As will be discussed later, the astronauts were hired early enough to participate in the design phases of Gemini and Apollo, and in my view this was one of the wisest decisions NASA made. A test pilot, more than any other type of aviator, must be objective.

Highlight (Yellow) | Location 562

It is all right for a squadron pilot to fall in love with his airplane; it is all he has to fly, and he might just as well enjoy it because it has already been designed; it exists in its present form and no one is going to change it now. He can afford to put on his blinders, compile a long list of prejudices in its favor, and develop an almost religious fervor when discussing it. He can look with scorn or pity on the unwashed from the neighboring squadron who are not so privileged. The test pilot cannot fall into this trap. Just because he has spent years flying Convair products doesn't mean that Lockheed's system is not just as good. He must learn that the Convair's delicate feel is great for an all-weather interceptor at high altitude, but that Lockheed's heavier stick forces are much more practical when working close to the ground.

Highlight (Yellow) | Location 568

Lindbergh's Spirit of St. Louis, for instance, had a longitudinal instability which generally would have been considered objectionable. However, Lindbergh didn't mind, because the constant attention he had to pay to the elevator control helped to keep him awake during the long lonely hours over the Atlantic.

Highlight (Yellow) | Location 586

a precise reading of the amount of kinetic energy (one half the mass times the velocity squared)

Highlight (Yellow) | Location 604

His boss barely looks up from the acey-deucey board. "Forgot to drop the tail hook, Collins; test is rescheduled for the morning." Collins never forgot the tail hook again.

Highlight (Yellow) | Location 620

There had also been extensive security and morals checks, "background investigations" as they are called, which had gotten all skeletons out of their closets, although I recall Al Shepard's lopsided, malevolent grin a few years later. "I still have a few secrets." They were la crème de la crème, and the nation loved them.

Highlight (Yellow) | Location 623

Some of them were disgruntled because they had not been selected (too old, too tall, no degree, or simply rejected), but others had actively avoided the selection like the plague. In retrospect, I can't say whether their reasons were obtuse or perceptive, but they do seem to me to be a fascinating indication of the time and place. Man, they were here to fly, not to be locked up in a can and shot around the world like ammunition. They were master craftsmen, artists, they were Jonathan Livingston Seagulls—they flew, in smooth control, in command. They flew day after day, in various machines to prove various things, but they flew. In Project Mercury, on the other hand, one rode; granted there had been only one Mercury flight so far, Al Shepard's fifteen-minute

Highlight (Yellow) | Location 632

who were all these people anyway? Slayton was the only one who had ever been assigned to Fighter Ops (Cooper had been in Engineering, for God's sake), and old Deke must have taken leave of his senses to forsake a fighter test assignment to get shot off somewhere in a tin can.

Highlight (Yellow) | Location 637

For me, the clincher came on February 20, 1962, with John Glenn's magnificent three-orbit flight aboard Friendship 7. Imagine being able to circle the globe once each ninety minutes, high above all clouds and turbulence!

Highlight (Yellow) | Location 640

in April 1962, NASA announced that it would select a new group of astronauts to supplement the Immortal Seven. NASA's terse announcement was encouraging, in that the agency was sticking with test pilots. This narrowed the field considerably, as there just weren't that many certified test pilots around, even though NASA was opening the selection to civilian as well as military pilots. A degree in one of the biological sciences or engineering was required, and candidates could not be more than six feet tall or over thirty-five years old, as of the day of selection.

Highlight (Yellow) | Location 652

We were told how to talk, dress, stand, and sit (wearing knee-length socks, of course, so no gross expanse of hairy leg intruded upon whatever Ph.D.'s in education think about), how to answer questions (not too long, not too short), how to drink at parties (long drink, make it last).

Highlight (Yellow) | Location 655

But, in my opinion, the apogee of the course was surely reached the day we learned how to hold our hands on our hips (doesn't anyone learn anything at home any more?). Thumbs forward, ladies! Thumbs to the rear, gentlemen! To reverse would not only be gauche (perhaps even worse) but, of course, would result in instant detection and rejection by NASA, the old meanies.

Highlight (Yellow) | Location 718

of the Original Seven, I had now met all but Gordo Cooper and Wally Schirra. They were an impressive bunch, photogenic, personable, with quiet assurance their common denominator. They somehow stood out in a crowd; if they had been chosen by a public-relations man, he could not have made a better choice. Yet I guessed that PR had had little to do with it. The emphasis in their selection was reported to have been on breadth of test pilot experience and on physical conditioning. As NASA gained experience, these two factors decreased somewhat in importance and were replaced by an emphasis on youth, education, and scientific specialties. But in the early days, the environment of space was expected to produce a variety of surprises, and who better to cope with them than a tough old bird who had been around the block once or twice. I think it was good reasoning.

Highlight (Yellow) and Note | Location 747

The impression that you made on our Selection Committee was favorable. Overall, however, we did not feel that your qualifications met the special requirements of the astronaut program as well as those of some of the other outstanding candidates. The special requirements, eh? Well, if I didn't have them, could I get them? Or was the door irrevocably closed behind me? Being a perpetual optimist, I focused on two points: first, there were strong rumors it wouldn't be more than a year or so before the burgeoning space program required selection of a third NASA astronaut group; and second, Conrad and Lovell were rejects from the Original Seven's screening process. Therefore, since I felt my basic problem was lack of experience, hopefully I could grow into a successful candidacy next time around. In the meantime, my job at Fighter Test, which had seemed the best in the world, no longer quite had a larger-than-life quality. I knew people in Houston and I envied them.

He had a very specific goal and amazing drive coupled with optimism.

Highlight (Yellow) | Location 794

got it going as fast as was legal (Mach 2), and then pulled back on the stick and zoomed upward as high as it would go. Trading kinetic energy for potential, up we would go, ever more slowly, until we floated over the top of a lazy arc in a not-so-bad simulation of the weightlessness of space. Using this technique, we would stagger up to around ninety thousand feet (I think Jim Roman held the record in our class, to the chagrin of us old test pilots). At ninety thousand feet, the sky overhead is so dark blue as to be almost the pure black of space; the weather, clouds, indeed practically all the atmosphere is below you; the horizon becomes a more sharply defined line, and you can detect the curvature of the earth. So near yet so far away! Grab a quick glimpse, then back to mundane chores in the cockpit, for the zoom up into this very thin air has not been without casualties. First, the afterburner portion of the engine blew out routinely at around sixty thousand feet. Next, the main-engine temperature would begin to climb dangerously, so that if the engine did not flame out of its own accord, it had to be shut down, usually in the vicinity of seventy-five thousand feet. Third, as soon as the engine quit, cabin pressurization was lost, and the only thing between the pilot and a blood-bubbling death was the pressure suit, inside which a safe if somewhat reduced pressure was maintained. Since pressure-suit gloves had been known to pop off, we were especially careful to check the wrist locks prior to these zoom flights, and then we would tape the locks down in place with adhesive. Space-age baling wire!

Highlight (Yellow) | Location 809

We derived all the equations necessary to prove to the most hardened skeptic that a spacecraft could orbit the earth; the higher the slower, the lower the faster. Right? Right—if at one hundred miles, one orbit takes an hour and a half; if we go up to 22,300 miles, one orbit takes twenty-four hours. So what? Well, the earth also takes twenty-four hours to go around once, so that the 22,300-mile satellite will stay over the same spot on the ground, i.e. be synchronous with the earth's rotation, which is fiendishly clever if you want to use it as a communications-relay satellite. Interesting applications of boring equations.

Highlight (Yellow) | Location 815

On the other hand, some of the space rules were the very antithesis of airplane rules. Consider the rendezvous problem, with one vehicle behind another and the laggard wishing to catch up. In an airplane, the answer is

quick and simple: use increased engine thrust to increase speed and approach from the rear; then reduce relative velocity to zero to fly in formation with the target. Try this in a spacecraft and it simply will not work: adding thrust toward the target puts one into a higher orbit, and high orbits are slower, remember, so instead of catching up, one falls behind. No, against all instincts, one must apply thrust away from the direction of the target, drop down into a lower, faster orbit, and then transfer back up into the original orbit at precisely the right point in the catch-up trajectory.

Highlight (Yellow) | Location 974

A gentleman of 32 who could calculate an eclipse, survey an estate, tie an artery, plan an edifice, try a cause, break a horse, dance a minuet and play the violin. —James Parton, *Life of Jefferson*

Highlight (Yellow) | Location 985

These planes were used to maintain our “proficiency”—that elusive quantity and quality which says that one is at home in the air (at least in a familiar airplane), that one is not going to be airsick, or become distracted by a strange new environment, or panic when some emergency arises. How real, or valuable, this “proficiency” is I do not know, but I do know that there is a world of difference between flying a simulator and flying a real machine. I think the key to it is that one can only go so far in pretending that life and limb are at stake. In the middle of the fanciest simulation, one can nip out to drink a cup of coffee or answer the telephone, but if a fog bank rolls in off the Gulf of Mexico and causes Ellington to go below landing minimums as you begin your approach, then the adrenalin pumps, the abdominal muscles tighten up, the pulse rate increases, and life or death decisions get made, well or poorly, but they get made. Even in a battered old T-33.

Highlight (Yellow) | Location 1088

I felt that the stresses on my family deriving from my weird occupation were sufficient without exposing them to a constant stream of well-meaning but insensitive reporters who would exacerbate the doubts and fears which most certainly were there, as they would be in the minds of any normal, sentient wife and offspring. Even more fundamental than that, I felt that I was paid to do my job to the best of my ability, and I put in long hours toward that end; but when I came home and that front door clicked behind me, that was it. My family life was my business and mine alone, and if I chose to open that door, it was perfectly proper to expect extra compensation for it.

Highlight (Yellow) | Location 1183

Mike Collins O.K. if you’re looking for a handball game, but otherwise nothing special. Lazy (in this group of overachievers, at least), frequently ineffectual, detached, waits for happenings instead of causing them. Balances this with generally good judgment and a broader point of view than most.

Highlight (Yellow) | Location 1221

Eminent scientists, like Tommy Gold of Cornell, fretted over a possible dust layer on the surface of the moon whose thickness might just exceed the height of the lunar module (LM). Others postulated a charge of static

electricity on the LM which would cause whatever dust was there to become attracted, and adhere, to the lander, obscuring the astronauts' view out of the windows. Lunar soil was thought by some to contain pure metallic elements which, when introduced on dirty boots into the pure oxygen in the LM cabin, would spontaneously burst into flame. Meteorites were known to exist in space, and with no atmosphere to protect it, the moon would clearly be an inviting target for them. Who knew how many there might be, or at what rate they might be impacting its surface? One has only to look at the moon's craters with naked eye to realize meteorites were not just a theoretical menace, but how real a threat to man they would be no one knew. At least the moon was well past the earth's Van Allen belts, which promised a healthy dose of radiation to those who passed and a lethal dose to those who stayed. Or how about a solar flare, that sporadic belch of energy spewing unpredictably from the sun?

Highlight (Yellow) | Location 1230

more worrisome were some of the unknowns involving transporting men and machines across a quarter of a million miles of hostile vacuum. The round trip would take about eight days, which, of course, meant that problems must not demand solutions in less than half that time.

Highlight (Yellow) | Location 1251

In regard to Gemini, for example, they had a theory that if an astronaut spent a couple of days in space and then was subjected to the sudden stress of re-entry into the earth's atmosphere, his autonomic nervous system would forget to perform all its vital functions, and among other snafus, it would allow blood to pool in his lower extremities and cause him to pass out because of insufficient blood to the brain. Even worse, once passed out, the astronauts would remain strapped in their seats, with heads up and feet down, so that the blood would never return to their upper body and they would die! Living and working with these people was like having an aunt who lives in a haunted house, or a close friend who sincerely believes in astrology and can't stop talking about it, especially delighting in reading you your horoscope on bad days. None of it is to be believed, but it's pretty goddamned difficult to ignore.

Highlight (Yellow) | Location 1259

The next big chance was the Skylab program, but everyone made it up there for nearly three months, so how about an eighteen-month round trip to Mars? Dr. Charles Berry, the biggest blabbermouth in the space program, has already said that an all-male crew might become too testy for this task, so the medical community is again abuzz with speculation over the psychological, if not the physical, barriers to space flight. The truth of the matter is that the space program would be precisely where it is today had medical participation in it been zero, or perhaps it would be even a little bit ahead, because we could have done without all the impedimenta and medical claptrap, such as blood-pressure cuffs, exercise ergometers, and urine-output measuring devices. All they did was add weight, and complexity, and rob time and energy from tasks of greater value.

Highlight (Yellow) | Location 1268

What would the temperature be inside the spacecraft during the constant sunlight on the way to the moon? With the sunny side baking and the shady side freezing, what equilibrium conditions would there be inside, where the

softies lived? The electronics might endure variations of a couple of hundred degrees, but the human body demands ridiculously narrow limits of temperature and pressure.

Highlight (Yellow) | Location 1273

And how about the guidance system on board the spacecraft? Could it really find its way to that magic spot in the sky where the moon would be three days after launch, and then navigate back to earth again? The allowable limits were spectacularly small. On the return trip, the atmospheric “re-entry corridor,” or zone of survivability, or whatever you wanted to call it, was only forty miles thick, and hitting a forty-mile target from 230,000 miles is like trying to split a human hair with a razor blade thrown from a distance of twenty feet.

Highlight (Yellow) | Location 1277

Granted, the primary responsibility for keeping the razor blade aimed precisely toward the absolute center of the hair would be the job of powerful ground tracking radars, coupled to gigantic computer complexes. But suppose the spacecraft experienced a communications failure and was unable to receive steering instructions from the ground. Would the astronauts be able to take over this delicate and vital assignment? The key to it would involve measuring the angle between a selected star and the moon’s or earth’s horizon, but how accurately could this be done? Assumptions were being made and equipment was being designed, but suppose mistakes were being built into the system? Suppose the sextant design was too crude for the precision required, or suppose the instrument became warped slightly by virtue of the fact that it was built on the ground (one G) but had to perform in space (zero G)? For that matter, was the earth’s horizon that clearly defined when seen from above the atmosphere? What would you see, the terrain horizon, or the atmospheric horizon, or some line in between? Perhaps it would shimmer, or move or vary with the angle of the sun’s rays, or perhaps a hundred other unknown factors might creep into the equation. Suppose all was known to a fare-thee-well, but then some absent-minded technician at Cape Kennedy loaded the wrong number for the earth’s diameter into the spacecraft computer, or, more likely, suppose the crew procedures incident to taking such a measurement were so involved and complicated that the astronauts made mistakes. Suppose each individual system aboard the spacecraft was a design masterpiece in itself, but when all the systems were added together they required twenty-six hours a day of crew attention? Suppose—suppose—suppose.

Highlight (Yellow) | Location 1290

I am merely trying to convey the flavor of MSC in 1964, when we had a mandate to fly to the moon but few hard facts with which to work.

Highlight (Yellow) | Location 1299

No test pilot in his right mind would, for instance, go faster than the speed of sound on his first flight in a new airplane: aerodynamics are quite different on each side of the sound barrier, and the handling qualities of an airplane are sorely tested in the highly sensitive region near Mach 1. But consider the Gemini spacecraft, mounted on top of a Titan II rocket. When it goes, it goes! All the way to orbit, banging through the speed of sound, from step A to step Z with no intermediate stops, with no opportunity to reconsider or redesign along the way. In the case of the Gemini-Titan, it was possible to run two unmanned test flights before putting Gus

Grissom and John Young aboard the third one; in the case of the Saturn V, again two unmanned flights preceded the Borman-Lovell-Anders flight of Apollo 8.

Highlight (Yellow) | Location 1306

the Saturn V was truly a new beast, far heavier and more powerful than any flying machine the world had ever seen, and it was deemed sufficiently trustworthy to send men to the moon on only its third flight.

Highlight (Yellow) | Location 1314

But if 1964 was for astronauts a year of technological doubt, a year for questioning scientists, engineers, and flight planners, it was also a year of great spiritual exuberance. We had not been with the space program long enough to become jaded by constant close contact with its marvels.

Highlight (Yellow) | Location 1319

Wise or not, we were on our way, and I remember 1964 as a vintner might: “Young, still tasting of tannin, rough around the edges, but solid, with body, a spicy bouquet, and a strength suggesting greatness in five years’ time.”

Highlight (Yellow) | Location 1361

Who would suspect that (Fe²⁺, Mg) Ti₂O₅ would be discovered at Tranquility Base in 1969 and that this new mineral would be called “armalcolite,” a name derived from the initial letters of Armstrong, Aldrin, and Collins.

Highlight (Yellow) | Location 1374

My own thinking was especially shortsighted in that it stopped abruptly at the first lunar landing. I thought that scores of preparatory flights would be required, and that the first landing would signal success and the end of the program, rather than the beginning of lunar exploration, with subsequent long-duration traverses complete with lunar rovers and scientific instrument packages. On the other hand, I believed (and still do) that lunar bases would one day be established and that man would fly to Mars. But these efforts would not be Apollo; President Kennedy had said, “... to land a man on the moon and return him safely to earth.”

Highlight (Yellow) | Location 1399

NASA-ese is no worse than Air Force-ese or State Department-ese; I suppose each has its place, although none of them seems a desirable substitute for English. Examine the sentence “Jones and Smith don’t get along well” translated into: (1) Air Force-ese: “It is considered that effective utilization of the potential allegiance between Jones and Smith is not being harmoniously exploited”; (2) NASA-ese: “The interface between Jones and Smith has gone divergent”; (3) State Department-ese: “Messrs. Jones and Smith’s abrasiveness vis-à-vis each other is counterproductive to their bilateral relationship; each considers the other a *bête noir*.” NASA-ese is not longwinded—it is just different, and in addition there are some no-nos in its vocabulary whose use marks one as a gauche outsider.

Highlight (Yellow) | Location 1408

Another thing one must learn is the proper use of units: spacecraft velocity, for example, is measured in feet per second, not miles per hour (or furlongs per fortnight, which is probably as sensible a unit as feet per second). One must learn the language if one is to study a foreign culture, and the language of space flight is as complex and precise as the maneuvers it describes, a fact that dismays NASA public-affairs officials at flight time

Highlight (Yellow) | Location 1412

Perhaps public support might have been stronger had the public really understood what the astronauts were talking about; on the other hand, I suppose some mystique is desirable in any enterprise.

Highlight (Yellow) | Location 1449

Our trip down into the canyon took nearly the whole day, so we spent the night in a charming inn at the bottom, and the next morning those of us who wanted to rented burros for an expedited ascent. I chose to ride, but picked an animal which stopped walking whenever I stopped kicking, so I got as much exercise as if I had been afoot. I also had plenty of time to contemplate the rapid pace at which I was speeding toward the moon. From supersonic jets at Edwards, I had progressed all the way to kicking a burro up out of the Grand Canyon. Just as the jets had their tricks, so did this creature; it seemed I was perpetually more involved with the transportation system than with the use to which it could be put. A scientist I would never be, at least not a geologist, but perhaps a transportation technologist deserved a role somewhere.

Highlight (Yellow) | Location 1482

Our bible was Air Force Manual 64-5, entitled Survival, and it makes jolly good bedtime reading. In fact, I have kept two copies of this little gold mine to this day, for one never knows. The manual opens on a cheery note: "Anything that creeps, crawls, swims, or flies is a possible source of food." Then it gets a bit too specific for my taste. "People eat grasshoppers, hairless caterpillars, wood-boring beetle larvae and pupae, ant eggs, and termites." Not me, babe! Oh yeah? Read on. "You have probably eaten insects as contaminants in flour, corn meal, rice, beans, fruits, and greens of your daily food, and in stores in general."

Highlight (Yellow) | Location 1531

Was not the moon staying forever one step ahead of us, just as described in one of Zeno's famous paradoxes: you see a turtle in the distance moving slowly ahead and you race to catch up with it. However, when you reach place A, where the turtle was when you first saw it, the turtle will have moved on to B. By the time you reach B, the turtle will have gone on to C. Therefore, no matter how many times the process is repeated, the turtle will always be ahead of you, and you can never quite catch up.

Highlight (Yellow) | Location 1540

It turns out that those little whitish disks you find in your heart-of-palm salad are part of a largish cylinder of edible stalk buried inside the trunk of certain varieties of palm tree. The trick is to be an expert palm-tree

identifier, because it takes a couple of hours to chop down a palm tree with just a small machete, and to whack through the tough fibrous exterior of the upper trunk to expose what could be either a delicious tender heart or more of the same inedible woody pulp, depending on the tree.

Highlight (Yellow) | Location 1556

Although threatening in appearance, prehistoric and dragon-like, iguanas are actually sluggish, harmless fellows, and deserve better treatment.

Highlight (Yellow) | Location 1565

We also met some Choco Indians, by arrangement, and were most impressed by their chieftain, Antonio, a man who appeared much younger than his years (forty?) because of his superb physical condition and unlined face. He also had a great impassive dignity, and seemed surprised at nothing, not even the news that we were practicing in his jungle so as to be able to fly to the moon. Perhaps he did not believe it. Years later, in Washington, I met Antonio again, at the Smithsonian, in the hall containing the original Wright Brothers airplane and, almost directly below it, the Apollo 11 command module. I patiently explained to Antonio about Apollo 11, about flying to the moon, and I learned from his interpreter that he was not particularly interested in the Apollo 11 craft. He didn't disbelieve my story, he simply could not relate it to the ugly triangular chunk of heat-shield-covered machinery. He liked the idea of flight to the moon, and obviously knew a lot about it—that it was a satellite of earth, and so forth—but the machinery he was interested in was the Wright Flier. This he could understand, the mannequin lying prone on the lower wing, and the crude engine powering a pair of wooden propellers. Here was useful technology.

Highlight (Yellow) | Location 1575

along the way I picked up a couple of hundred companions, chiggers, evenly distributed from the waist down. I cannot adequately explain to the unchiggered what they are missing. My dictionary says simply that chiggers are “the parasitic larva of certain mites.” It doesn't say they are also abominable little red creatures who burrow into your skin, where they ultimately die. Uneasy in their terminal tunnels, they either jump about or dig deeper, or secrete some irritant, or something; at any rate they itch like crazy.

Highlight (Yellow) | Location 1609

The trick here is to get up off the ground a few inches if possible, as it can be 30 degrees cooler a foot above the ground than it is on the surface, and to keep the sides of your shelter or tent or whatever open, to allow free circulation of air.

Highlight (Yellow) | Location 1616

Neil Armstrong and Dave Scott brought Gemini 8 down suddenly and unexpectedly in the Pacific Ocean rather than the Atlantic, but they were picked up by a destroyer stationed there for that contingency purpose. Scott Carpenter's Aurora 7 overshot its landing area by 250 miles, but a helicopter quickly reached him. Neither crew was in the water over three hours. Thus, as in the case of so much astronaut training, the information has never

really been needed, but it nonetheless was prudent to try to prepare people for as many variations upon the expected theme as possible.

Highlight (Yellow) | Location 1620

I think this has been one of NASA's great strengths in the conduct of the Gemini, and, even more so, the Apollo series. Large groups of capable men have sat for years around conference tables with engineering schematic drawings and asked themselves, "What happens if?" In some few cases the answer is simply "Tough," there is no out, no solution, no alternate course. But these are few and far between.

Highlight (Yellow) | Location 1645

strictly from a personal point of view, there is a good deal of satisfaction to be derived from these "feel so good when they stop" situations.

Highlight (Yellow) | Location 1646

One cannot really appreciate being chigger-free without first sampling chiggers. Living in temperature-controlled, water-tight enclosures, with food and drink instantly available, causes us to forget—or worse yet, never to know—the pleasure of simply feeling warm and dry and full.

Highlight (Yellow) | Location 1669

As Deke Slayton delicately put it in a memo: "... Nothing engenders hatred more rapidly than having an astronaut appear at the Podunk Center Elks Club on his own after having officially turned down the local congressman's request for a Chamber of Commerce or Rotary Club dinner. The message is: No astronaut has authority to accept speaking engagements or public appearances ... without approval from higher authority. When in doubt, call..."

Highlight (Yellow) | Location 1699

Rule 2 is to stop worrying about what you say, it really doesn't matter. This rule applies especially when talking to the press out in the hustings. I always had a secret fear, especially in the early days, that the shallowness of my technical knowledge would be exposed. "Tell me, Astronaut Collins, what will be the eccentric anomaly of that orbit, to three decimal places?" in sonorous tones. "I'm sorry, sir, I don't know," in a small piping reply. This reverse Mittyism is ridiculous—all the guy wants is some little thing that will make both of you look good, and if you're misquoted, who reads the West Wissioming Daily Astonisher, anyway?

Highlight (Yellow) | Location 1703

Rule 3 is don't blow your cool. I used to get angry at being shoved, or asked to sign fourteen autographs by one lady who pushed in front of a nice kid who had been waiting patiently for one, or at seeing the schedule slip further and further behind. Don't bother. Detach yourself. Repeat a favorite Omar Khayyam quatrain over and over. I suggest "The moving finger writes..." Pretend you are not down on the ballroom floor, surrounded, but are up on a chandelier, swinging gently, looking down at the bizarre scene below with detached amusement.

Highlight (Yellow) | Location 1731

finally, after much bombast, Bryan was allowed to proceed. When it was all over, two old farmers assessed the situation. “He were really sumpin’, weren’t he?” “Yes, sir, and that feller what came after weren’t no slouch either.” While my introducer was droning away, interrupted only by periodic squeals from the PA system, I felt a persistent pulling at my pant leg. I didn’t want to kick a Boy Scout in the head,

Highlight (Yellow) | Location 1746

Since most of us had been involved, as test pilots, in seeing new machines grow from drawing board to flight line, hopefully we would be able to transfer our knowledge to this new realm, to look with jaundiced eye at what the theoreticians had put together, to call upon the strong points of some similar programs and to avoid the pitfalls of others. This at least was the theory, and I think it worked well.

Highlight (Yellow) | Location 1752

and I had indicated a desire to work on the development of the pressure suit, or space suit, as the public calls it.

Highlight (Yellow) | Location 1757

to really understand guidance and navigation, one must dig into its attendant mathematical base, and this jungle is full of ominous creatures with names like vector and tensor.¹ Therefore, I was pleased when Al issued a memo announcing our assignments: Buzz Aldrin—mission planning Bill Anders—ECS, radiation and thermal Charlie Bassett—training and simulators, operations handbooks Al Bean—recovery systems Gene Cernan—spacecraft propulsion and Agena Roger Chaffee—communications, DSIF Mike Collins—pressure suits and EVA Walt Cunningham—electrical and sequential, non-flight experiments Donn Eisele—attitude and translation controls Ted Freeman—boosters Dick Gordon—cockpit integration Rusty Schweickart—future programs and in-flight experiments Dave Scott—guidance and navigation (G&N) C. C. Williams—range operations and crew safety

Highlight (Yellow) | Location 1779

Buzz Aldrin—mission planning This involved going to an interminable series of meetings at MSC, attempting to define how many flights would be required to get to the moon and how they should be organized for maximum efficiency. The rendezvous problem was probably the biggest single unknown from the crew point of view, and there were almost endless variations in technique for bringing two vehicles together in earth or lunar orbit. Should the interceptor approach from above or below, in dazzling sunlight or in inky black, at a great overtaking speed or very slowly? These were not philosophical questions, but practical ones which by proper analysis could be converted into gallons of fuel remaining, or mathematical formulae describing probability of success.

Highlight (Yellow) | Location 1790

It was a sensible use of Buzz’s education, which included a doctoral dissertation on orbital rendezvous.

Highlight (Yellow) | Location 1791

Bill Anders—ECS, radiation and thermal ECS stands for environmental control system, alias the plumber's delight. A baffling array of pipes, tanks, valves, connectors, switches, filters, heaters, fans, sensors, etc., whose details almost obscured its central purpose, that of providing an environment of 100 percent oxygen at a pressure of five pounds per square inch (to prevent body fluids from boiling in the vacuum of space). Temperature and humidity limits also had to be controlled, so that the crew could not only breathe but live comfortably for the fourteen days or so they would be up. The system also had to cope with the carbon dioxide which humans exhale, and arrange for potable water to be stored or produced.

Highlight (Yellow) | Location 1796

While the ECS determined the environment within, and was a hardware design problem, radiation and thermal refer to the outside environment and are less concerned with hardware than with limitations on its use. The Apollo command and service module (CSM), for instance, has severe thermal constraints. By that I mean, if you hold the CSM steady in any one position in space for a long time, the up-sun side gets too hot and the down-sun side too cold. Fluids within start boiling and freezing, and all kinds of bad things happen.

Highlight (Yellow) | Location 1807

In many cases it is more difficult to simulate something than it is to do it, and in some ways it was more difficult to get the simulators "flying" than the real McCoys which followed a couple of years later. In this kind of situation, it is vital to have someone with balanced judgment, who has flown a variety of new machines, to say, "Yes, we must have this effect" or "No, we can get by without that one."

Highlight (Yellow) | Location 1820

Gene Cernan—spacecraft propulsion and Agena Agena was the auxiliary engine which was launched unmanned and whose power was later used by a Gemini which had docked with it to push the two vehicles together into a higher orbit than would have been possible using Gemini's energy alone.

Highlight (Yellow) | Location 1826

Gene's master's degree in aeronautical engineering concentrated on rocket-engine technology, which made him especially well suited to the job.

Highlight (Yellow) | Location 1827

Roger Chaffee—communications, DSIF DSIF stands for deep space instrumentation facility. The DSIF consisted of tracking stations around the world,

Highlight (Yellow) | Location 1830

As early as 1963, the DSIF had bounced radar signals off the planet Mercury, over 60 million miles away, so we weren't too concerned about its power, but its accuracy was of vital concern to those of us who would be

navigating with its help. On board the spacecraft, communications consisted of a wide array of transmitters and receivers, covering various portions of the energy spectrum from low frequency to the superduper high-frequency range called S-band.

Highlight (Yellow) | Location 1846

The electrical system is only slightly less boring; at least it includes the fuel cells, which miraculously generate electrical power by combining oxygen and hydrogen, getting water as a by-product. It's the opposite of the old high-school science experiment, wherein an electrical current is passed through water (H₂O), causing it to separate into a bell jar of H and a bell jar of O.

Highlight (Yellow) | Location 1852

Donn Eisele—attitude and translation controls

Highlight (Yellow) | Location 1855

Attitude really means which way something is pointed. Up, down, toward the sun, or what? Remember doing an aileron roll in a T-38 a couple of chapters ago? If done properly, this roll would result in no change in pitch or yaw attitude (the nose would stay fixed on a point), but the craft would change 360 degrees in roll attitude. Translation, on the other hand, means moving through space, be it up or down, left or right. If you throw this book against the far wall, and I wouldn't blame you at this stage, you would have caused it to translate laterally or longitudinally (however you define it) that distance, and then translate vertically down to the floor. Inside a spacecraft, one flies with both hands. The left grasps a translational hand controller, the right an attitude hand controller. Between the two, all maneuvers are possible.

Highlight (Yellow) | Location 1872

can arrange the switches in such a fashion that when he removes his hand from the right-hand stick he commands a zero rate, and the spacecraft will stay fixed in inertial space, gyro-stabilized, firing its thrusters as necessary to continue pointing in exactly the same direction, within the limits of the selected dead band. If the switches are not so selected, the unattended spacecraft will wander off and point wherever the laws of physics say it should. The first scheme is better for control, but at the expense of fuel consumption;

Highlight (Yellow) | Location 1879

Ted Freeman—boosters Boosters have several aliases: launch vehicles, rockets, missiles.

Highlight (Yellow) | Location 1882

historically they are of supreme importance because without them there would be no space age. Tsiolkovsky, Goddard, Von Braun—these chaps never dreamed about simulators or recovery systems or mission planning, but about the raw rocket power which would be required to put men into orbit, onto the moon, out beyond Jupiter, or wherever. In my own view, the pendulum has swung; rockets are about as interesting as the powder

which propels a bullet. To be fascinated by them is to prove McLuhan's point that the medium is the message, or perhaps it is a more Freudian thing.

Highlight (Yellow) | Location 1887

The boundaries of leaving or staying are generally defined by two sets of limits: if it wanders off past a certain angle, which might cause you to point in the wrong direction, or if it changes direction at an unacceptable rate, indicating a predilection to tumble ass over tea kettle. So maximum angles and rates are analyzed and memorized by the crew, which generally has time to respond to such excursions. On the other hand, if the thing blows up, it usually does so with an instantaneous finality which makes crew action futile.

Highlight (Yellow) | Location 1935

Heart of the Apollo system is a lump of metal slightly larger than a basketball, called the inertial measuring unit, or IMU. The IMU contains three gyroscopes mounted at right angles to each other and connected to the spacecraft by gimbals. When the gyroscopes are spinning at full speed, they keep the IMU pointed in one direction ("fixed in inertial space"). Think of it as a "stable table" around which the spacecraft turns. Now if we knew which way the IMU was pointed, in comparison to the stars for example, then we could measure which way the spacecraft was pointed, relative to the same stars, by measuring the angles between the spacecraft axes and the IMU axes. These three angles (pitch, yaw, and roll gimbal angles) tell you which way you are pointed, relative to the stars, which are fixed in inertial space. Gyroscopes precess, or wander off, so the IMU must be restored to precise stellar alignment periodically by the astronaut, who selects two stars, peers at them through a telescope or sextant, superimposes cross hairs on them (one at a time, of course), and then pushes a button at the instant of perfect alignment. He then tells his computer which two stars (by pre-stored numbers or celestial coordinates) he has selected. Then the IMU is torqued around to its correct new alignment. All this keeps your attitude information up to date, but does not help in telling you where you are, only which way you are pointed.

Highlight (Yellow) | Location 1946

The where part comes from a mathematical thing called a state vector, which is stored in your computer. A state vector has seven parts: three tell you position, three velocity, and one time. Position is expressed by three components (x , y , and z) of distance from a reference point, while velocity ($\dot{x}$, $\dot{y}$, and $\dot{z}$) is the rate of change of those three distances. Time (t), of course, pinpoints the time at which you were at any one spot (x , y , z) heading which way ($\dot{x}$, $\dot{y}$, $\dot{z}$). The computer knows where it was when it started (Launch Pad 39 at Cape Kennedy) and keeps book on all subsequent travel. During times of engine firing, accelerometers are turned on to record changes in velocity, but in coasting flight the computer keeps book by an involved mathematical computation of the gravitational pull exerted by sun, earth, and moon upon the spacecraft. As time goes by, the state vector loses accuracy and it must be updated, either by the ground computers, which can radio a new state vector up to the spacecraft, or by the astronauts, who can measure the angle between a selected star and the earth or moon horizon. If we want to change course, it's as simple as ABC. We just get Roger Chaffee's tracking network to send up a state vector and a couple of other things to Dave Scott's computer, which feeds pointing commands through Dick Gordon's instrument panel to

Donn Eisele's controls, which cause Walt Cunningham's electricity to power Gene Cernan's engines, which fire, to get us out of Bill Anders's radiation zone into the position called for by Buzz Aldrin's flight plan. The rest of you guys must be loafing!

Highlight (Yellow) | Location 1981

I could get a lot done in Houston, working with NASA engineers in the crew systems division, but most of the serious design reviews and equipment inspections took place at the contractors' plants, which meant my old T-33 was kept busy jumping from Connecticut to California to Delaware to Texas to Massachusetts, trying to keep up with developments. And developments were not all rosy, especially in the Apollo suit department. In the first place, there is a kind of love-hate relationship between an astronaut and his pressure suit. Love because it is an intimate garment protecting him twenty-four hours a day, hate because it can be extremely uncomfortable and cumbersome.

Highlight (Yellow) | Location 2008

it is necessary to explain what a pressure suit is designed to do, and why that is so difficult. First and foremost, it must be airtight, so that it can be pumped up ("pressurized") to protect the astronaut from the vacuum of space. Without gas pressure around him, he would have nothing to breathe, and even if you solved this by giving him an oxygen mask, he would still die quickly because the fluids inside his body would vaporize—his blood would literally bubble. So we must begin with a gas bag which can be pressurized to around 3.7 pounds per square inch, which is about what you need if you use 100 percent oxygen as we did.⁶ The gas bag, or bladder as it was called, was made of thin neoprene rubber, contoured to fit the body. The bladder material was soft and pliable, moving easily with the astronaut, provided the suit was not pressurized. When it was pumped up to 3.7 psi, it was a different matter. The next time you stop by your neighborhood filling station, watch them pump up an inner tube. It comes out of its box soft and floppy, but as the air pressure inside it increases, it becomes a fairly rigid doughnut. Same thing with a pressure suit bladder, even though we are only pumping to 3.7 psi instead of the inner tube's higher pressure. To prevent ballooning of the inner tube, it is restrained inside a relatively inflexible tire casing; in similar fashion, a pressure suit bladder is contained by a "restraint layer." Here is where black magic enters the rational world of design. The restraint system must allow the pressurized suit to bend as the astronaut bends, twist as he twists, and in general act as a tough extra layer of skin. It's not too difficult to design a flexible joint for a knee which only moves back and forth in the same plane, but how about a shoulder joint? It is capable of incredibly complex motions, and the suit must be able to follow them all, without undue effort by its occupant, without leaking, without changing shape, without zinging back to some awkward position the instant the astronaut relaxes.

Highlight (Yellow) | Location 2048

If things seem to be getting complicated, remember that all we have done so far is to allow a man to operate in a vacuum. Now we must add the other hazards of that vacuum: the blinding energy of the sun, the subfreezing temperature of any object shielded from the sun, and the possible penetration of the suit by tiny projectiles called micrometeorites. Fortunately, heat, cold, and micrometeorites could all have a common solution—a thick cover layer which would be a superb thermal insulator and an adequate shield against high-velocity impacts. The thermal part of the problem was straightforward in that we had definite design inputs: the temperature in

darkness was expected to be—250 degrees F, in direct sunlight +250 degrees F, and at the bottom of a lunar crater + 310 degrees F. Hence the thermal limits of the Gemini cover layer (−250 degrees F to +250 degrees F) and the Apollo cover layer (−250 degrees F to +310 degrees F) were well defined; it was simply a search for the most efficient materials, which turned out to be multiple layers of thin Mylar. The micrometeorite problem was different. Micrometeorites are the sharks of outer space: normally absent, usually harmless when present, but on rare occasions terribly destructive.

Highlight (Yellow) | Location 2058

Our mathematical model showed very simply that if you stayed out in space a long time, your chances were excellent of getting hit by a small micrometeorite and poor of getting hit by a large one. In this kind of situation it is impossible to design 100 percent protection; common sense must take over. Since Mylar was not a good energy absorber, we added one layer of a felt-like material, which in conjunction with a tough nylon outer covering gave us good protection against the small, dust-like particles which make up the vast majority of the micrometeorite flux. At first we did not know how to package this thermal and micrometeorite protection. We tried overcoats and capes and other schemes, and finally decided to build the protection directly into the suit cover layer.

Highlight (Yellow) | Location 2115

Flying in space turned out to be easier than flying in the zero-G airplane. It was also a lot more pleasant. The zero-G airplane was a converted Air Force KC-135 (essentially a Boeing 707) which had had all the seats removed from the cabin and the interior padded. It reminded me strongly of a padded cell. The airplane usually flew out of Wright-Patterson AFB in Dayton, Ohio, although it could be summoned on special occasions to Houston or the Cape. It could imitate weightlessness for slightly over twenty seconds at a crack by flying parabolic arcs. The pilots would dive it down steeply, pull up fairly sharply (2 or 3 Gs), and then push the airplane over the top at exactly the right rate to keep our bodies suspended between ceiling and floor in the aft cabin. At least that was the theory. In practice, even the smoothest pilot wiggled a little bit, and we tended to bang around in the back.

Highlight (Yellow) | Location 2124

The climax of the flight was a carefully staged act featuring Ernie Volgenau, one of our instructors. A collegiate wrestling champion, Ernie looked exactly like Clark Kent, and his wife had sewn—especially for this occasion—a Superman suit for him to wear. It was made out of long-john underwear dyed sky blue, with brief red shorts over it, the great Superman S insignia on the chest, and a long flowing cape. Two of the biggest, fattest students (since they would weigh nothing, it didn't matter) were picked to be “criminals,” and as soon as the pilot got us all weightless on the next parabola, Ernie uttered a scream of triumph, picked up one crook with each hand, examined them with disgust as he held them at arms' length, then knocked their terrified heads together, and hurled them the length of the fuselage! Talk about high camp, and all of it recorded for posterity by movie cameras.

Highlight (Yellow) | Location 2135

people would always get sick and throw up, usually right at the beginning of each flight. There is something very unsettling, even for experienced aviators, about repeating forty or fifty parabolas, the body alternating constantly between zero and 2 Gs. A typical planeload included cameramen, engineers, and various supporting personnel—most of whom were not accustomed to this kind of thing and who were almost sure bets to throw up within the first half dozen parabolas. That didn't help. Some astronauts threw up. I never quite did, but I was close enough at times to be utterly miserable. I found that it helped to minimize my head movements, but frequently that was impossible as we thrashed about trying to retrieve make-believe micrometeorite packages from make-believe Agenas, or whatever the task of the day was. The strange thing about the zero-G airplane was that instead of becoming acclimated to it, my tolerance seemed to decrease. Toward the end of my Gemini career, one had only to say "KC-135" to make me feel quite queasy. Thank God, we didn't use it much for Apollo crew training.

Highlight (Yellow) | Location 2166

After these episodes I considered going to Deke, confessing all, and leaving the program. But of course, I didn't, because I wanted desperately to stay, and because I felt I could talk myself out of this ridiculous condition. After all, I had worn pressure suits on a dozen occasions at Edwards, including the F-104 zoom flights, and had never had the slightest tendency to feel trapped inside them. Furthermore, the claustrophobia, or whatever, was an infrequent visitor, and I could sail through most tests with no difficulty. Granted it would take only one such incident in flight to put me and my fellow crew members in grave peril, but on the other hand, on about half the planned flights the pressure suit would not be used at all, except in the highly unlikely event of cabin pressure failure. Somehow I could, I would, overcome this absurd impediment to flying in space. As time went by, things got better. I learned to request maximum cooling flow at the beginning of a test sequence, even if it meant I half froze. I also learned to concentrate on my breathing, and could convince myself that all was well, that I really was getting enough to breathe. Finally, as I got to know the equipment better, and as the really tough tasks were re-engineered to make them simpler, I found fewer and fewer occasions when I would be gasping for breath. I never did discuss the matter with anyone, so I don't know whether I alone had the problem, but I must say I've never seen or heard of another claustrophobic astronaut. I must also say that my own love-hate relationship for my pressure suits is more strongly developed because of all this. I did have a highly successful space walk on Gemini 10, wearing pressure suit number G-4C-36. It was a beautiful suit, with special modifications to give me extra visibility and mobility; it fit perfectly, and I felt very much at home in it. Its paper-thin bladder, glued together with consummate care by some nice ladies in Worcester, was all that kept my 3.7 psi oxygen from escaping into the zero psi infinity surrounding me as I dangled from the end of my fifty-foot tether. For G-4C-36 I felt as close to love as one can feel for metal, fabric, and rubber; but when I think generically of pressure suits, I think, Ugh, nasty little coffins. I'd be pleased never to get inside one again.

Highlight (Yellow) | Location 2279

When the fighter pilot "pulls positive Gs," as he calls it when he pulls out of a dive, that is "eyeballs down" in astronaut parlance. A pilot's "negative G" is "eyeballs up." Too many positive Gs and the blood cannot reach your brain and you "black out." Too many negative Gs and too much blood is forced into your brain and you "red out," a much more hazardous condition. The astronaut works at 90 degrees to all this. Unlike a pilot's seat,

his couch is oriented in such a way that all his Gs are transverse, or “eyeballs in.” While “eyeballs in” can hurt your chest and impair your breathing, you can endure much higher levels without losing vision or consciousness.

Highlight (Yellow) | Location 2295

On Apollo 17, for example, Gene Cernan spent seven and a half hours covering some twelve miles, statistics which would simply not have been believable in 1965. At that time, some astronauts were even suggesting that development of the back pack be abandoned, that instead we should simply plan to work with umbilical lines from the LM, which would have limited our radius of action to fifty feet or so. The reasoning behind this was that the back packs were becoming larger, heavier, and more complex; that they were making a small cockpit even more crowded; that they would never be as reliable as umbilicals; that their weight meant less fuel could be carried (and the fuel budget was very tight); and that all we needed from the moon was a few rocks, anyway. My own opinion was that we should continue working on the fancy gear and postpone decisions on its use until we saw how it developed. The fact that this was also an expensive course to follow did not worry me a bit. One nice thing about Apollo was that no one ever told us we were running the price up too high.

Highlight (Yellow) | Location 2316

the Molly Brown plopped into the Atlantic some sixty miles short of target after three hectic orbits. Gus had time to change orbits three times by firing his thrusters (a necessary prelude to rendezvous), and John had time to run through a variety of subsystem tests. He also managed to eat a corned-beef sandwich which Wally Schirra had caused to be placed in his pressure suit pocket—a fact unknown to NASA, which reacted somewhat hysterically. The medics claimed that somehow that sandwich had negated the flight’s medical protocol, while the engineers claimed that crumbs from it could easily have invaded the guts of the machinery with catastrophic effect. Some members of Congress became apoplectic, charging NASA with having lost all control of the astronaut group. I think most of us could have strangled Wally for bringing a tornado of upper-echelon attention down on us because of something as trivial as a corned-beef sandwich. Deke was even moved to write us all a memo: “... the attempt ... to bootleg any item on board not approved by me will result in appropriate disciplinary action. In addition to jeopardizing your personal careers, it must be recognized that seemingly insignificant items can and have affected the prerogatives of follow-on crews. Witnes [sic] this memorandum.” It was also about this time that the practice of naming spacecraft was directed to cease,² a disappointment to Jim McDivitt and Ed White, who wanted to name their upcoming Gemini 4 American Eagle.
